Section 13.4: Electricity Generation

Mini Investigation: What Factors Affect Electricity Generation?, page 602

A. Answers may vary. Sample answers:
Moving the magnet more quickly into the coil increased the amount of current produced.
B. Reversing the pole of the magnet inserted in the coil did not affect the amount of current produced.
C. Using two magnets greatly increased the amount of current produced.
D. Increasing the number of windings in the coil increased the amount of current produced. Decreasing the number of windings in the coil decreased the amount of current produced.

Research This: Wind Turbines, page 604

For large-scale land-based turbines:
A. Answers may vary. Sample answer:
The turbine spins at a rate of about 30 to 60 rotations per minute. The electricity is controlled through a gearbox, which is connected between the shaft from the turbine and the shaft driving the generator. The gearbox controls the rate of rotation of the shaft driving the generator, which determines how much electricity is generated.
B. Answers may vary. Sample answer:
The wind turbine must be regularly inspected and maintained to ensure proper operation of the controller, the computer that starts, stops, and rotates the turbine appropriately. The controller also controls the generator, the rotor (which consists of the blades of the turbine and the hub where they connect), and other electronic and mechanical parts of the turbine.
C. Answers may vary. Students' pamphlets should include information on the development of their chosen wind turbine technology and a summary of how the technology works.

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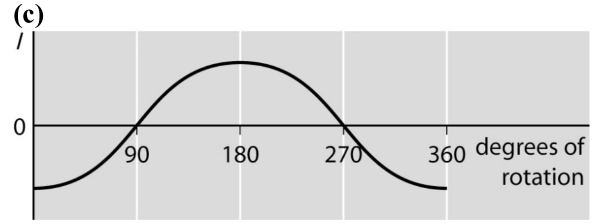
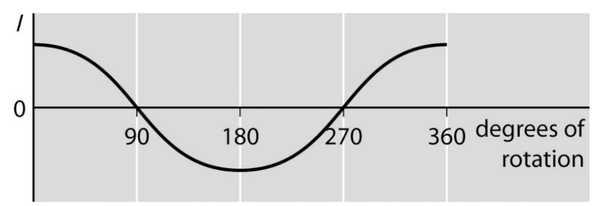
1. Diagrams should look like Figure 1 from page 599 of the Student Book with the arrow at the axis of rotation indicating a counterclockwise rotation instead of clockwise.
(a) The current will be at its maximum at the angles of 0° and 180° relative to the starting point, because the plane of the loop will be parallel to the external magnetic field at these angles.
(b) The current will be zero at the angles of 90° and 270° relative to the starting point, because the plane of the loop will be perpendicular to the external magnetic field at these angles.
(c)



(d) Reversing the polarity of the external magnets would reverse the initial direction of the current. The graph from part (c) would look like the following:



2. The rotation rate of the loop in the generator in Figure 1 is the same as the frequency of the alternating current that it generates. Each time the loop rotates through half of a complete rotation, the current changes direction.
3. The frequency of alternating current in North America is 60 Hz, so a generator armature in North America must rotate 60 times per second.
4. As the shaded side of the armature moves away from the north pole of the external magnet, Lenz's law determines the left side of the armature to be a south magnetic pole. Using the right-hand rule for a coil, the direction of the conventional current is down across the front of the coil. Looking at the diagram and following the conventional current around the coil, it can be seen that the conventional current exits the coil through the inner split ring and travels through the external circuit in the counterclockwise direction.